Health Assessment and Promotion

Exam Two Worksheet Answers and Sources

Answers follow the 34 numbered items in the updated NURS 3120 worksheet. Each answer includes a source locator for review. PDF page numbers refer to the page position in the file, which may differ from the printed slide number. Source filenames are clickable.

Chapter 15 Head Neck and Lymphatics

1 Compare hypothyroidism and hyperthyroidism

Feature	Hypothyroidism	Hyperthyroidism
Metabolism	Deficient thyroid hormone; decreased metabolism.	Excess thyroxine; increased metabolism.
Weight	Weight gain.	Weight loss.
Heart rate	Bradycardia.	Tachycardia; palpitations may occur.
Temperature tolerance	Cold intolerance; feels extremely cold.	Heat intolerance; excessive sweating.
Skin and appearance	Pale, dry skin; thin, brittle nails; puffy face.	Excessive sweating; bulging or puffy eyes; thyroid enlargement.
Energy and other symptoms	Fatigue, muscle aches, constipation, irritability and depression.	Fatigue, nervousness and diarrhea.
Causes in the course material	Thyroid hormone deficiency; the notes identify Hashimoto disease.	Graves disease, thyroid nodules and thyroiditis.

Source: Chapter 15 8th edition.pdf, PDF pp. 8 and 14, health history and altered thyroid signs and symptoms

Additional hypothyroid findings in the quiz: Dry, coarse, sparse hair; edema around the eyes; a puffy, dull face. Exophthalmos is not a hypothyroid finding.

Source: assessment exam 2 quiz 1.pdf, PDF p. 7, hypothyroidism select all that apply item

2 Lymph node findings and palpation technique

Normal: Nodes may be nonpalpable. If palpable, they are small, soft, movable, discrete and nontender; most are less than 1 cm long. Discrete means individually separate and distinct.

Abnormal: Enlarged, tender or painful nodes suggest acute infection. Enlarged nodes without pain can occur with cancer metastasis. Findings that differ from the expected soft, movable, discrete pattern require further assessment.

Technique: Use gentle circular motions with the pads of the fingers. Palpate both sides simultaneously to compare symmetry. Assess the preauricular, postauricular, occipital, tonsillar, submandibular and submental nodes, and continue through the head and neck drainage areas shown in the course diagram.

Source: Chapter 15 8th edition.pdf, PDF pp. 6–7 and 13, lymph nodes and palpation; assessment exam 2 quiz 1.pdf, PDF p. 9, normal cervical node documentation item

3 Five traumatic brain injury prevention strategies

1. Wear a seat belt whenever driving or riding in a motor vehicle.

2. Wear a helmet or appropriate protective headgear for sports and recreation, including bicycle riding.

3. Prevent falls by removing loose rugs, installing bathroom support handles and reviewing medicines that cause dizziness or sleepiness.

4. Never drive under the influence of alcohol or drugs.

5. Protect children with stair safety gates, window guards and soft material beneath playground equipment.

Source: Chapter 15 8th edition.pdf, PDF p. 24, TBI client education; assessment exam 2 quiz 1.pdf, PDF p. 11, helmet teaching item

4 Compare the five headache types

Terminology: The course slides describe these as headache types; sinus, cluster, tension and medication overuse headaches are separate from migraine.

Sinus: Deep, constant, throbbing pain and pressure across or beneath the eyes. It may be tender and extend across the face. Often occurs with a cold or acute sinusitis and worsens with movement or bending forward.

Source: Chapter 15 8th edition.pdf, PDF p. 17, sinus headache

Cluster: Sudden, intense, stabbing pain, often with tearing, eyelid drooping or a runny nose. The quiz describes severe unilateral pain around one eye. Alcohol may precipitate it. It is more common in people assigned male at birth and may recur seasonally in clusters separated by months or years.

Source: Chapter 15 8th edition.pdf, PDF p. 18, cluster headache; assessment exam 2 quiz 1.pdf, PDF p. 1, headache comparison item

Tension: The most common type; dull, aching tightness in frontal, temporal or occipital regions. The quiz describes a bilateral tight band across the forehead and temples. Associated with inadequate sleep, stress, anxiety or depression; duration varies.

Source: Chapter 15 8th edition.pdf, PDF p. 19, tension headache; assessment exam 2 quiz 1.pdf, PDF p. 1, headache comparison item

Migraine: Severe, throbbing pain, usually unilateral, that interferes with usual activities and may last up to three days. Nausea, vomiting and sensitivity to light, sound or smells are common. Prodromal symptoms may include visual changes, vertigo, tinnitus or tingling. Triggers include stress, emotional events, anxiety and foods such as cheese, chocolate or alcohol; a genetic cause is discussed.

Source: Chapter 15 8th edition.pdf, PDF p. 20, migraine headache

Medication overuse: Rebound headache associated with large doses of pain relievers for at least three months. Dull, constant pain occurs most days; location and intensity vary. Nausea, vomiting, diarrhea, disturbed sleep and restlessness may accompany it.

Source: Chapter 15 8th edition.pdf, PDF p. 21, medication overuse headache

5 Subjective questions for headache assessment

Use the course's COLDSPA pain assessment approach. Ask what the headache feels like, when it began, where it occurs, how long it lasts, how severe it is, its pattern or frequency, what brings it on or worsens it, and what symptoms accompany it. Ask about nausea, vomiting, light or sound sensitivity, visual changes, dizziness, tinnitus, tingling, nasal symptoms, stress, sleep and food or alcohol triggers. Ask about head injury, medications and frequency of pain-reliever use. These questions apply the headache characteristics and triggers described in the slides.

Headache diary: Encourage documentation of frequency, duration and potential triggers to identify contributing factors.

Source: Chapter 15 8th edition.pdf, PDF p. 8, COLDSPA health history; PDF pp. 17–21, headache characteristics; Chapter 25 8th edition.pdf, PDF p. 9, headache history using COLDSPA; assessment exam 2 quiz 1.pdf, PDF p. 10, headache diary item

6 Next step after an abnormal thyroid assessment

If the thyroid is enlarged, auscultate the thyroid lobes for a bruit. A normal thyroid has no audible bruit. A soft, blowing or swishing bruit reflects increased blood flow and is associated with hyperthyroidism. This is the next physical assessment step specified in the chapter.

Source: Chapter 15 8th edition.pdf, PDF p. 12, palpation and auscultation; PDF p. 25, thyroid review question

Chapter 16 Eyes

7 Eye assessment techniques

Snellen chart: Have the client stand 20 feet from the chart and cover one eye. Ask the client to read the smallest full line possible; assess the other eye in the same manner. Record the result as 20/20 or the appropriate acuity notation. The first number is the testing distance; the second corresponds to the last full line read.

Source: Chapter 16 8th edition.pdf, PDF p. 16, distant visual acuity; Chapter 16 — Full Timestamped Transcript.pdf, PDF p. 23, 00:24:56–00:25:56

PERRLA and pupillary light reflex: Inspect pupil size, equality and roundness; compare with a pupil gauge as needed. Shine a penlight into an eye and assess immediate constriction in that eye (direct response) and the opposite eye (consensual response). Assess both eyes. Normal pupils are equal, round and reactive to light and accommodation; they constrict together. Fixed, unequal or nonreactive pupils are abnormal.

Source: Chapter 16 8th edition.pdf, PDF pp. 13–15, pupil inspection, light reflex and PERRLA; Chapter 25 8th edition.pdf, PDF p. 14, pupil abnormalities

Six cardinal fields of gaze: Keep the client's head straight and ask the client to follow a finger or target with the eyes through the six positions: right, upper right, lower right, left, upper left and lower left. Observe for smooth, symmetric movement. Jerky oscillation or nystagmus and limited movement are abnormal. This tests extraocular muscle function and cranial nerves III, IV and VI.

Source: Chapter 16 8th edition.pdf, PDF p. 17, six cardinal fields of gaze; Chapter 16 — Full Timestamped Transcript.pdf, PDF pp. 24–25, 00:26:04–00:28:07

Confrontation field test: Position yourself about two feet from the client at eye level. The client covers one eye while you cover the opposite eye, and the uncovered eyes maintain direct gaze. Move an extended finger into view from the periphery and ask the client to report when it is seen. Compare with when you see it; test all four visual quadrants and repeat for the other eye. Seeing the target at the same time is expected. Delayed or absent detection suggests peripheral field loss.

Source: Chapter 16 8th edition.pdf, PDF p. 18, confrontation field test; Chapter 16 — Full Timestamped Transcript.pdf, PDF pp. 26–27, 00:28:09–00:30:30

Accommodation: Have the client shift focus from a distant point to a nearby finger, pencil or other target. Observe for both pupils to constrict and the eyes to converge toward the near object. Normal accommodation permits near focusing through increased lens curvature.

Source: Chapter 16 8th edition.pdf, PDF pp. 7 and 15, accommodation and PERRLA

8 Cataracts glaucoma and macular degeneration

Cataracts: Gradual clouding of the lens, commonly after age 65. Findings include blurred vision, fading or yellowing colors, glare or light sensitivity, halos and double vision. Risks include age, smoking, diabetes, excessive alcohol, sunlight exposure, hypertension, obesity, previous eye injury, inflammation or surgery, and prolonged corticosteroid use.

Source: Chapter 16 8th edition.pdf, PDF p. 23, cataracts; assessment exam 2 quiz 1.pdf, PDF pp. 6 and 10, cataract risks and eye disorder matching

Glaucoma: Optic nerve damage associated with increased intraocular pressure; loss of peripheral vision and possible halos. Damage is irreversible, and there may be no early warning signs. Risks listed include age over 40, family history, high intraocular pressure, East Asian ancestry, farsightedness, previous eye injury or surgery, diabetes, hypertension and corticosteroid use. The slides also emphasize earlier onset, faster progression and higher risk in people of African descent.

Source: Chapter 16 8th edition.pdf, PDF pp. 11 and 24, glaucoma risks and characteristics

Macular degeneration: Damage to the macula causes loss of sharp central vision, with blind spots or blurred/distorted straight-ahead vision. Risks include increasing age, White race, smoking, female sex, obesity, prolonged sunlight exposure, light eye color, diets high in fat/cholesterol or sugar, hypertension, cardiovascular disease and inactivity.

Source: Chapter 16 8th edition.pdf, PDF p. 22, macular degeneration

9 Eye terminology

Term	Definition
Myopia	Impaired far vision or nearsightedness.

Term	Definition
Diplopia	Double vision.
Photophobia	Inability to tolerate light.
Presbyopia	Age-related impaired near vision, typically after age 45.
Scotoma	A blind spot, surrounded by normal or slightly diminished peripheral vision.
Strabismus	Constant malalignment of the eyes.
Nystagmus	Oscillating or shaking movement of the eyes.

Source: Chapter 16 8th edition.pdf, PDF pp. 9, 17 and 20, subjective findings and terminology

Chapter 17 Ears

10 Six tinnitus risk factors or associated conditions

a. Cerumen impaction.

b. Middle ear infection.

c. Hypertension.

d. Hearing loss.

e. Environmental noise exposure, including occupational noise.

f. Ototoxic medications such as aspirin, gentamicin, tobramycin, furosemide, naproxen and vancomycin.

Source: Chapter 17 8th edition.pdf, PDF pp. 7–8, tinnitus history and terms

11 Normal and abnormal tympanic membrane findings

Normal: Pearly gray, shiny, translucent and essentially flat with a slight central concavity; no abnormal bulging or retraction. Visible landmarks and a cone of light are expected.

Abnormal finding	Likely association
Red, bulging membrane; distorted, diminished or absent light reflex	Acute otitis media.
Yellowish, bulging membrane with bubbles behind it	Serous otitis media.
Bluish or dark red color behind the membrane	Blood behind the eardrum after skull trauma.
White spots	Scarring from infection.
Perforation	Trauma associated with infection.

Source: Chapter 17 8th edition.pdf, PDF pp. 3 and 10, tympanic membrane; assessment exam 2 quiz 1.pdf, PDF pp. 4–5, tympanic membrane matching and healthy ear item

12 Six hearing loss prevention strategies

a. Avoid or reduce exposure to loud occupational and recreational sounds.

b. Wear hearing protectors when loud sound exposure cannot be avoided.

c. Immunize children against childhood diseases linked to hearing loss.

d. Be immunized against rubella before pregnancy or during childbearing years.

e. Avoid ototoxic medications when possible and review medication-related hearing risk with the prescriber.

f. Do not place foreign bodies in the ears.

Source: Chapter 17 8th edition.pdf, PDF pp. 14–15, hearing loss prevention

13 Older adult hearing and presbycusis

Age-related ear changes: Ear hair becomes coarse and wiry; cilia become coarse and stiff. Cerumen may accumulate or become impacted even though production decreases, causing dryness. The tympanic membrane may be whiter and duller. Consider the effect of hearing loss on communication, following instructions and hearing alarms or warnings.

Presbycusis: Gradual sensorineural hearing loss from degeneration in the inner ear. High-frequency sounds are affected; consonants are harder to hear than vowels, words may sound garbled, and female voices can be harder to hear. It is common with aging but remains an abnormal hearing finding requiring assessment.

Source: Chapter 17 8th edition.pdf, PDF pp. 12 and 14, older adult considerations and communication safety; assessment exam 2 quiz 1.pdf, PDF p. 8, presbycusis item

Chapter 18 Nose Mouth and Throat

14 Normal tonsils and priority abnormalities

Assessment and normal findings: With the mouth open, use a tongue depressor and penlight to inspect tonsil color, size, exudate and lesions. Tonsils may be absent following tonsillectomy. When present, they should be pink, symmetric and generally 1+, without swelling, exudate or lesions.

Grading: 1+ = visible; 2+ = halfway between the tonsillar pillars and uvula; 3+ = touching the uvula; 4+ = touching each other.

Priority abnormalities: Marked enlargement, swelling, exudate, lesions or a tonsillar abscess. The priority is maintaining the airway, particularly when enlarged tonsils approach or touch one another.

Source: Chapter 18 8th edition.pdf, PDF pp. 10 and 16, tonsil examination, airway priority and tonsillar abscess

15 Palpable sinuses and palpation technique

Palpable: Frontal sinuses above the eyes and maxillary sinuses in the upper jaw/cheek region are accessible to examination.

Nonpalpable: Ethmoid and sphenoid sinuses are located too deep in the skull for direct palpation.

Technique: Gently palpate the frontal and maxillary regions for tenderness. Normal sinuses are not tender; tenderness may indicate infection. The quiz identifies gentle palpation of the maxillary and frontal areas as the first sinus assessment action for headache and facial pressure.

Source: Chapter 18 8th edition.pdf, PDF pp. 4 and 12–13, sinus anatomy and assessment; assessment exam 2 quiz 1.pdf, PDF p. 4, sinus assessment item

16 Four sinusitis teaching considerations

a. Reduce respiratory infection exposure through good hand hygiene and a yearly influenza vaccine.

b. Avoid known allergy triggers.

c. Stay hydrated.

d. Inhale steam or rinse the nose regularly with saline solution.

Source: Chapter 18 8th edition.pdf, PDF p. 14, sinusitis education

17 Normal and abnormal tongue findings

Normal: Pink, moist, moderate-sized and midline, with many papillae and no lesions. The underside is smooth, shiny and pink with visible veins. Symmetric, smooth movement and equal strength are expected.

Abnormal appearance: A swollen, smooth, beefy-red tongue with few or no papillae is associated with vitamin B12 or niacin deficiency. White curd-like patches that scrape off and leave reddened, easily bleeding mucosa suggest thrush. Raised white feathery areas that cannot be scraped off suggest oral hairy leukoplakia in immunocompromised clients. Black hairy tongue is associated in the slides with antibiotics or Pepto-Bismol, smoking and alcohol. Fissures may be harmless; deep longitudinal fissures can occur with dehydration. Inspect lesions particularly on the lateral tongue and underneath it because these are cancer sites emphasized in the notes.

Abnormal movement: Tongue atrophy, fasciculations, weakness or deviation to the affected side can indicate cranial nerve XII dysfunction.

Source: Chapter 18 8th edition.pdf, PDF pp. 8–9 and 15–17, tongue findings; Chapter 25 8th edition.pdf, PDF p. 20, cranial nerve XII

18 Seven groups of oral cancer risk factors

a. Tobacco use, including cigarettes, pipes, cigars and smokeless tobacco.

b. Heavy alcohol consumption.

c. HPV infection.

d. Prolonged sunlight exposure and fair skin.

e. Age over 45 and being assigned male at birth.

f. Poor oral hygiene and poor diet/nutrition.

g. Weakened immune system and marijuana use.

Source: Chapter 18 8th edition.pdf, PDF pp. 5–6 and 18–19, oropharyngeal cancer risks

Chapter 19 Respiratory

19 Locations of normal breath sounds

Sound	Normal location
Bronchial	Over the trachea and larynx.
Bronchovesicular	Over the major bronchi: anterior upper sternum in the first and second intercostal spaces; posteriorly around the scapular region.
Vesicular	Over the peripheral lung fields.

Source: Chapter 19 8th edition-1.pdf, PDF p. 20, normal breath sounds; assessment exam 2 quiz 1.pdf, PDF p. 9, normal breath sound matching

20 COPD findings and breathing education

Signs and symptoms: Dyspnea, tachypnea, air trapping, barrel chest, tripod positioning, pursed-lip breathing, diminished breath sounds, wheezes or crackles, accessory muscle use, cyanosis and clubbing. COPD includes emphysema and chronic bronchitis and makes exhalation difficult because airflow out of the lungs is obstructed.

Source: Chapter 19 8th edition-1.pdf, PDF pp. 7, 26 and 28, air trapping and COPD findings

Pursed-lip breathing: Teach slow exhalation through pursed lips and prolong the out-breath. This slows respirations, improves expiration, delays small airway collapse and reduces air trapping.

Position and energy conservation: Sit or stand leaning forward with the arms supported (tripod position) to stabilize the chest and reduce the work of breathing. Support the arms during ADLs if raising them worsens breathlessness. Pace activities, take frequent rests, use assistive devices and divide tasks into smaller steps. Avoid smoking and use protective measures for occupational respiratory irritants.

Source: Chapter 19 8th edition-1.pdf, PDF p. 27, COPD client teaching; assessment exam 2 quiz 1.pdf, PDF pp. 3 and 11–12, pursed-lip breathing and tripod position items

21 Six groups of lung cancer risk factors

a. Cigarette smoking, the principal risk factor emphasized in the slides.

b. Secondhand smoke exposure.

c. Asbestos exposure.

d. Radon exposure.

e. Air pollution and vaping exposure.

f. Prior chemotherapy or radiation in smokers and a poor diet lacking fruits and vegetables.

Source: Chapter 19 8th edition-1.pdf, PDF p. 31, lung cancer risk factors

22 Proper lung auscultation technique

Use the diaphragm of the stethoscope directly on bare skin. Avoid listening over bone. Compare corresponding sites in a side-to-side pattern and assess all lung areas. Listen to at least one full inspiration and expiration at every location. Have the client breathe normally so the client does not hyperventilate; allow breaks as needed, particularly with asthma or COPD. Identify normal breath sounds and any adventitious sounds.

Source: Chapter 19 8th edition-1.pdf, PDF pp. 19, 23 and 25, auscultation method and guidelines; Fast facts Auscultation of lungs.png, Guidelines section

23 Respiratory terminology

Term	Definition from the course material
Dyspnea	Shortness of breath.
Orthopnea	Difficulty breathing while supine; may require sitting upright or sleeping with pillows.
Tachypnea	Rapid respiratory rate; the slides use more than 24 breaths/minute.
Bradypnea	Slow respiratory rate; the slides use fewer than 10 breaths/minute.
Apnea	Suspension or cessation of breathing.
Kussmaul respirations	Rapid, deep, labored breathing or air hunger; associated with diabetic ketoacidosis and expelling CO2.
Cheyne-Stokes respirations	Cycles of gradually increasing and decreasing breathing depth followed by apnea.
Air trapping	Retention of air because the client cannot exhale completely; associated with COPD.
Biot respirations	Irregular rate and depth followed by periods of apnea; associated with meningitis or severe brain damage.
Ataxic respirations	Markedly disorganized breathing with irregular and varying depths; described as a more extreme expression of Biot respirations.
Hypoventilation	Slow, shallow breathing with inadequate CO2 exchange, causing CO2 to remain in the bloodstream.

Source: Chapter 19 8th edition-1.pdf, PDF pp. 6–8 and 11, respiratory terms and orthopnea

24 Five COPD risk factors

a. Active smoking.

b. Secondhand smoke exposure.

c. Asthma.

d. Occupational exposure to dust and chemicals.

e. Exposure to fumes from burning fuels.

Source: Chapter 19 8th edition-1.pdf, PDF p. 27, COPD risks

Chapter 6 Mental Status

25 Delirium Alzheimer disease and depression

Delirium: Acute onset; lasts hours to weeks and fluctuates. Attention is impaired, recent and immediate memory are impaired, and thinking is disorganized. Arousal may fluctuate from lethargy to hypervigilance. Associated causes include drug toxicity or interactions, acute illness, trauma, worsening chronic illness, fever, UTI, dehydration and fluid/electrolyte disorders.

Alzheimer disease: Slow, insidious onset with progressive decline. Recent memory and the ability to learn new information worsen; managing finances may become difficult. Early confusion involves familiar places and objects. Later, the person may not recognize people or places, language deteriorates and voluntary activity is lost. Depression is common.

Depression: Often abrupt onset; lasts weeks to months or years. The client is generally alert and may be distractible, but attention is minimally impaired. Some memory remains intact. Thinking is intact but may center on helplessness or self-depreciation.

Source: Chapter 6 8th edition.pdf, PDF pp. 23 and 25, comparison and Alzheimer symptoms; assessment exam 2 quiz 1.pdf, PDF p. 8, inability to manage monthly bills item

26 Four groups of substance misuse risk factors

a. Beginning substance use at a young age; genetic or family predisposition.

b. Other mental health conditions, including depression, anxiety or PTSD, and risk-taking/thrill-seeking behavior.

c. Family trauma, abuse, neglect or parental substance use.

d. Community and societal influences, including peer pressure, prevalent substance use, violence, poverty, stigma and discrimination.

Additional risks listed: Sex assigned at birth and the pattern/type of substance use; opioids, nicotine, methamphetamine and intravenous use are highlighted as having stronger associations with dependency.

Source: Chapter 6 8th edition.pdf, PDF p. 10, substance use disorder risk factors

27 Levels of consciousness

Level	Definition and response
Alert	Fully aroused and aware of the environment. Alertness does not by itself guarantee correct orientation; a confused client can still be alert.
Lethargy	Slowed or sluggish speech, mental processes or motor activity; arouses to name.
Obtunded	Decreased awareness and response to the environment; may require repeated calling of the name and gentle shaking to arouse. May give one-word answers, follow simple commands and be disoriented.

Level	Definition and response
Stupor	Arouses only with vigorous stimulation; brief eye opening and minimal response, with confusion during arousal.
Coma	No spontaneous movement, no purposeful movement with stimulation and no verbalization.

Source: Chapter 6 8th edition.pdf, PDF p. 18, level of consciousness; Fast facts mental status.png, levels of consciousness section

Chapter 25 Neurologic Assessment

28 Cranial nerves I through XII

The chart separates all 12 cranial nerves. The course assesses III, IV and VI together for eye movements and IX and X together for swallowing, palate movement and gag response. Each row cites the relevant PDF page.

Cranial nerve	Assessment technique	Normal and abnormal findings
I Olfactory	With eyes closed, occlude one nostril at a time and ask the client to identify a familiar smell. Not routinely tested unless indicated by loss of smell, trauma or abnormal mental function.	Normal: Correctly identifies the odor. Abnormal: Cannot smell or identify it; may reflect nerve injury or frontal lobe pathology. Source: Chapter 25 8th edition.pdf, PDF p. 12.
II Optic	Test distance vision with Snellen and near vision with a handheld Jaeger chart; test visual fields by confrontation. Inspect the optic disc/fundus with an ophthalmoscope.	Normal: Expected visual acuity and fields; normal disc appearance and red reflex. Abnormal: Reduced vision, field deficits or papilledema; associated with retinal, optic nerve or brain pathology. Source: Chapter 25 8th edition.pdf, PDF p. 13; Chapter 16 8th edition.pdf, PDF pp. 16 and 18.
III Oculomotor	Assess lid elevation, six cardinal fields of gaze, convergence, and pupil responses to light and accommodation.	Normal: Smooth, coordinated movement; eyelid covers about 2 mm of iris; pupils constrict simultaneously. Abnormal: Ptosis, limited movement, or a unilaterally dilated pupil unresponsive to light/accommodation. Source: Chapter 25 8th edition.pdf, PDF p. 14.
IV Trochlear	Assess six cardinal fields of gaze, focusing on inferomedial eye movement.	Normal: Smooth, coordinated eye movement. Abnormal: Limited movement, weakness or paralysis affecting the expected direction of gaze. Source: Chapter 25 8th edition.pdf, PDF p. 14.
V Trigeminal	Palpate temporal and masseter muscles while the client clenches the teeth. Test sharp/dull and light touch over forehead, cheek and chin. Assess corneal reflex with a fine cotton wisp as described in the course.	Normal: Bilateral muscle contraction, correct sensory identification and bilateral blink. Abnormal: Weak/asymmetric jaw contraction, impaired facial sensation or absent corneal reflex. Source: Chapter 25 8th edition.pdf, PDF p. 15.

Cranial nerve	Assessment technique	Normal and abnormal findings
VI Abducens	Assess six cardinal fields of gaze, focusing on lateral eye movement.	Normal: Smooth, coordinated lateral gaze. Abnormal: Limited lateral movement or paralysis. Nystagmus is abnormal during the combined gaze examination. Source: Chapter 25 8th edition.pdf, PDF p. 14.
VII Facial	Inspect facial symmetry. Ask the client to raise eyebrows, frown, close eyes tightly against resistance, show teeth, smile and puff cheeks.	Normal: Symmetric facial movements. Abnormal: Weakness/asymmetry, inability to close the eye or wrinkle the forehead; may occur with Bell palsy or stroke. Sensory testing is not routine. Source: Chapter 25 8th edition.pdf, PDF p. 16.
VIII Vestibulocochlear or Acoustic	Assess hearing in each ear during normal conversation and with whispered words at 1–2 feet. Weber and Rinne tests are listed but are not emphasized for this course.	Normal: Hears whispered words. Abnormal: Reduced hearing. Inner ear vestibular dysfunction may also produce vertigo or imbalance. Source: Chapter 25 8th edition.pdf, PDF pp. 9 and 17; Chapter 17 8th edition.pdf, PDF p. 7.
IX Glossopharyngeal	Together with X, assess swallowing and the gag reflex; warn the client before touching the posterior pharynx with a tongue depressor. Observe soft palate/uvula movement while saying ah.	Normal: Swallows without difficulty; gag response is present. Abnormal: Dysphagia or reduced/absent gag response; the notes acknowledge that some clients have a reduced or absent gag reflex. Source: Chapter 25 8th edition.pdf, PDF p. 18.
X Vagus	Assess voice/hoarseness, swallowing and symmetric elevation of the soft palate and uvula when the client says ah. Assess gag reflex with IX.	Normal: Palate and uvula rise symmetrically; swallowing is easy. Abnormal: Hoarseness, dysphagia, failure of the palate to rise or absent gag response. Source: Chapter 25 8th edition.pdf, PDF p. 18.
XI Spinal accessory	Ask the client to shrug shoulders against your hands and turn the head to each side against resistance; assess trapezius and sternocleidomastoid contraction.	Normal: Strong, symmetric contraction. Abnormal: Shoulder droop, asymmetry, weakness or muscle atrophy. Source: Chapter 25 8th edition.pdf, PDF p. 19.
XII Hypoglossal	Ask the client to protrude the tongue, move it side-to-side against tongue-depressor resistance and return it to the mouth. Inspect symmetry and atrophy.	Normal: Smooth, symmetric movement and equal bilateral strength. Abnormal: Fasciculations, atrophy or tongue deviation toward the affected side. Source: Chapter 25 8th edition.pdf, PDF p. 20.

29 Modifiable and nonmodifiable stroke risks

Modifiable risks listed on the course slide: Hypertension, obesity, sedentary lifestyle, poor diet, smoking, diabetes mellitus and high cholesterol. Teaching targets include smoking cessation, control of blood pressure/diabetes/cholesterol, exercise, healthy weight, healthy diet and limiting alcohol.

Nonmodifiable risks listed on the course slide: Age; family history/genetics/AB blood type; race; ethnicity; sex; anxiety/depression; sleep apnea; and prolonged exposure to high levels of air pollution. This reproduces the slide's grouping, including the last three factors.

Source: Chapter 25 8th edition.pdf, PDF p. 34, stroke risk factor classification and teaching

Additional risks on the stroke fast-facts sheet: Heart disease including coronary disease, cardiomyopathy, heart failure and atrial fibrillation; brain aneurysms; inflammatory conditions or infections; personal stroke history; alcohol; illegal drugs; stress; and prolonged NSAID use such as ibuprofen or naproxen. The fast-facts sheet lists these risks without sorting them into modifiable/nonmodifiable categories.

Source: Fast facts Stroke.png, risk factors and education sections

30 Romberg test indication and positive result

Purpose: Assess balance and the ability to maintain posture without visual orienting information. It is part of the course's cerebellar/balance assessment and is relevant when the client reports imbalance or difficulty coordinating movement.

Technique: Have the client stand with feet together and arms at the sides. Note baseline steadiness with eyes open; then ask the client to close the eyes and maintain the position for about 20 seconds. Stand close enough to catch the client if balance is lost.

Normal: Maintains posture and balance; slight swaying can occur.

Positive: Loss of balance with eye closure, marked swaying, falling or widening the feet to prevent a fall. The course associates a positive finding with loss of proprioception, loss of vestibular function and cerebellar ataxia, including alcohol intoxication or multiple sclerosis.

Source: Chapter 25 8th edition.pdf, PDF pp. 9, 22 and 24, imbalance history and Romberg test

Chapter 24 Musculoskeletal

31 Normal wrist and elbow range of motion

Wrist: Flexion (bend the hand downward), extension (return toward straight), hyperextension, adduction, and radial/ulnar deviation (move the hand toward the thumb or little-finger side). Compare both wrists for symmetric, comfortable motion. Wrists should be nontender, without redness, swelling or nodules.

Elbow and forearm: Flexion (bend the forearm toward the body), extension (straighten it), supination (turn the palm upward) and pronation (turn the palm downward). Compare both sides in flexed and extended positions; motion should be symmetric without tenderness, swelling, redness or structural abnormality.

General normal ROM: Active joint movement produces no tenderness, pain or crepitation. If active ROM is limited, gently assess passive motion and measure with a goniometer if abnormal.

Source: Chapter 24 new curriculum.pdf, PDF pp. 10, 12 and 18–19, ROM and elbow/wrist assessment; Chapter 24 — Full Timestamped Transcript.pdf, PDF pp. 45–50, 00:48:40–00:54:07

32 Five groups of osteoporosis risk factors

a. Age 65 or older; being assigned female at birth; menopause, especially early menopause; White or Asian ethnicity.

b. Family history of osteoporosis/osteoporotic fracture, previous fracture or low bone mineral density.

c. Low BMI, inadequate calcium/protein intake, vitamin D deficiency or conditions impairing nutrient absorption, such as celiac or Crohn disease.

d. Alcohol intake, nicotine use, sedentary lifestyle or frequent falls.

e. Long-term glucocorticoid use such as prednisone, rheumatoid arthritis or hyperthyroidism.

Source: Chapter 24 new curriculum.pdf, PDF pp. 32–33, osteoporosis risk factors; assessment exam 2 quiz 1.pdf, PDF pp. 12–13, osteoporosis risk item and feedback

33 Abnormal findings of the feet and ankles

Sprain: The ankle is a common site of stretched or torn ligaments; pain and swelling are abnormal findings to assess.

Gouty arthritis: Tender, painful, red, hot and swollen metatarsophalangeal joint of the great toe.

Rheumatoid arthritis: Nodules may be palpable at the posterior ankle.

Hallux valgus and bunion: Great toe deviates laterally and may overlap the second toe; an enlarged, painful, inflamed bursa may form on the medial side.

Plantar fasciitis: Tenderness at the plantar surface of the calcaneus/heel due to irritation or inflammation of the supportive tissue in the arch.

Other abnormal findings: Tenderness, heat, swelling, nodules, bony prominences, crepitus or restricted motion. Expected movements are ankle plantar flexion/dorsiflexion and foot inversion/eversion.

Source: Chapter 24 new curriculum.pdf, PDF pp. 24–25, foot and ankle findings; Chapter 24 — Full Timestamped Transcript.pdf, PDF pp. 62–63, ankle sprains and foot findings; assessment exam 2 quiz 1.pdf, PDF p. 2, acute gout item

34 Five older adult musculoskeletal considerations

a. Reduced bone density increases fracture risk, particularly at the wrist, hip and vertebrae; osteoporosis or osteomalacia further increases risk.

b. Reduced flexibility can limit ROM and make some examination positions difficult.

c. Impaired position sense increases fall risk; assess gait, balance, mobility aids and safe performance of ADLs.

d. Kyphosis is common, sometimes with a compensatory backward head tilt.

e. A bow-legged appearance may develop when osteoarthritis wears down the medial knee cartilage.

Functional assessment: Apply ROM and strength findings to the client's ability to perform essential activities safely and independently. The timed up and go test evaluates rising from a chair, walking 10 feet, turning, returning and sitting; the course uses 12 seconds or longer as an older adult fall-risk threshold.

Source: Chapter 24 new curriculum.pdf, PDF pp. 26–27 and 39, older adult considerations, functional assessment and TUG; assessment exam 2 quiz 1.pdf, PDF p. 12, decreased position sense item